



EDU TERIA

72nd BPSC

Prelims Online Batch

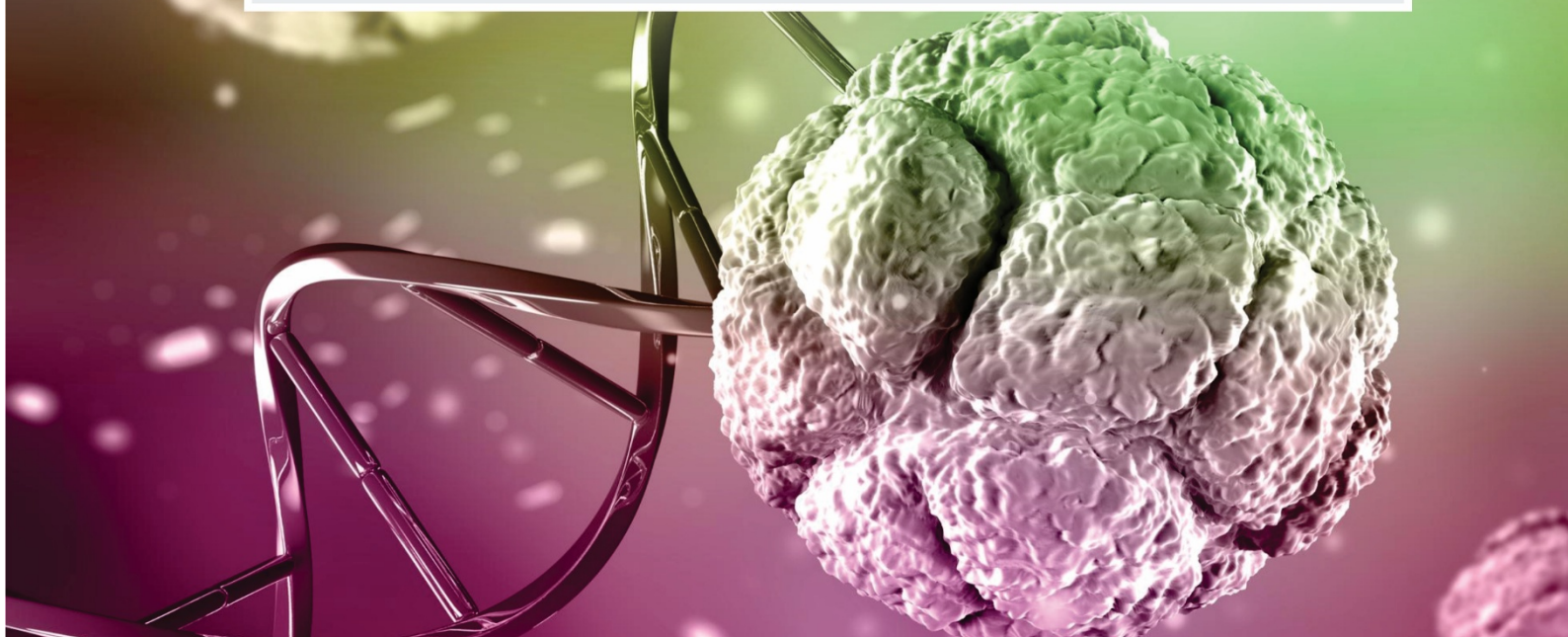
BIOLOGY

Topic wise
Notes

LECTURE

17

Circulatory System, Part - 03



Circulatory System, Part - 03

Heart Beat

Heartbeat is the rhythmic contraction and relaxation of the heart that pumps blood throughout the body.

Heart Beat—One contraction and relaxation combined called heart beat.

Adult Human

- Heart Rate $\rightarrow 72 / \text{min.}$
- New Born $\rightarrow 120\text{-}140/\text{min.}$
- Athletes $\rightarrow 130\text{-}140/\text{min.}$

Heart Sound

Heart sounds are produced by closing of heart valves during heartbeat. Can be heard by an instrument known as **Stethoscope**, which is based on reflection of sound wave. It was invented by **lennec**.



Types of Heart Sounds

Made up two types of voice notes.

Lub (S1)

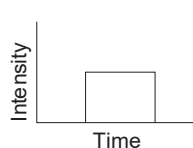
Caused by closure of AV valves
(Tricuspid + Mitral valve)

Dub (S2)

Caused by closure of Semilunar valves
(Aortic + Pulmonary valve)

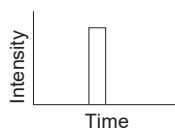
Heart Beat—due to reflection of sound wave.

(i) Lub



Dull & prolonged
(upto 0.15 sec)

(ii) Dub



Sharp and pointed
(upto 01 sec.)

Produced during ventricle produced during Systole. Ventricle diastole.

Events During

Cardiac Cycle—Beginning of one heart beat to the next heart known as cardiac cycle.

72 — 60s

1 — $60/72 = 0.8 \text{ sec}$

One heartbeat will take 0.8 sec.

- Ventricle systole - 0.35
- Ventricle Diastole - 0.55
- Atrium contract - 0.1 sec
- Atrium relax - 0.7 sec

Cardia output—Amount of blood pumped by heart in one minute known as cardiac output.

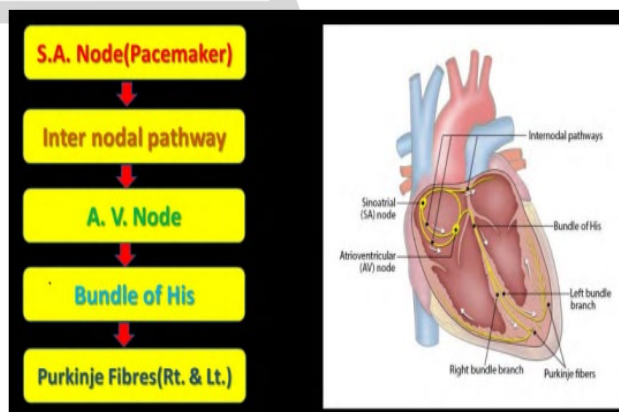
Value of normal cardiac output— 5 lit (min)

Stroke Volume— Amount of blood pumped by heart in its one heart beat

Normal Value— 70/min

Regulation of Cardiac Activity:

- The heart is myogenic, as its activity is regulated by nodal tissues.
- The heart's function is moderated through ANS (autonomic nervous system) by a neural centre in the medulla oblongata.
- Sympathetic neural signals and adrenal medullary hormones - increase heart rate and cardiac output.
- Parasympathetic neural signals - decrease heart rate.





Blood Pressure

Pressure exerted by blood in aortic wall measured as blood pressure. It is measured in two stages—

1. **Ventricle systolic pressure**—Pressure exerted by blood in aortic wall during ventricle systole. In adult humans it is measured in 120 mm of Hg.
 2. **Ventricle diastolic pressure**— Pressure exerted by blood on aortic wall during ventricle diastole. In healthy adult human it is measured in 80 mm Hg.
- Blood pressure is measured by sphygmomanometer.

Blood Pressure = 120(Systolic)/80 (Diastolic)

Note : Increase in heart rate—Tachycardia

Decrease in heart rate—Brady cardia

Electrocardiogram	Electrocardiograph
Graphical representation of electrical activity heart.	A Machine used to measure electrical activity of heart.

ECG:

The electrical activity of the cardiac cycle can be recorded in a graphical form by electrocardiogram or ECG.

- ❖ Various phases of each cardiac cycle are represented by a letter from P to T.
 - **P wave** → excitation or depolarisation of atria
 - **QRS complex** → depolarisation of ventricles
 - **T** → repolarisation of ventricles (excited to normal)
 - The no. of QRS complex recorded in a time period tells the heart rate of the patient. Any abnormality or disease can be diagnosed by ECG if the graph shows any deviation from a regular pattern.

Disorders of the Circulatory System:

- **High Blood Pressure (Hypertension):** The normal blood pressure of a human being is 120/80 mm Hg. 120 is the systolic pressure and 80 is the diastolic pressure. High bp > 140/90 mm Hg for a longer duration may lead to various heart diseases and may also adversely affect the brain and kidneys.

- **Hypotension :** It is also called low blood pressure. Hypotension or low blood pressure is the occurrence of persistent systolic
- arterial pressure of less than 110 mm Hg and diastolic arterial pressure of less than 70 mm Hg. It is caused by
- persistent vasodilation of arterioles, reduced ventricular pumping, valvular defects, anaemia and deficient diet.
- **Coronary Artery Disease (CAD):** It is also referred to as atherosclerosis. It affects the supply of blood to the heart. The lumen of the arteries become narrower due to deposition of cholesterol, fat, calcium or fibrous tissues.
- **Angina (Angina Pectoris):** The severe chest pain due to the scarcity of oxygen reaching the heart muscles, when there is interrupted blood flow.

Angina occurs due to :-

- (1) when enough oxygen is reaching to heart muscle
- (2) when not enough oxygen is reaching to heart muscle
- (3) the deposition of carbohydrates in artery
- (4) the deposition of protein in artery

Heart Failure: When the heart fails to pump enough blood to meet the requirement of the body. Lungs congestion is one of the symptoms of the disease. It is different from a heart attack, where there is sudden damage of heart muscle due to an insufficient blood supply or cardiac arrest when the heart stops beating.

Cardiac Arrest

Cardiac arrest is a sudden loss of heart function, where the heart stops beating effectively, leading to cessation of blood circulation.

It is a medical emergency and can cause death within minutes if not treated immediately.

Coronary Thrombosis: It is a condition when there is a clot formation in the coronary artery. It happens frequently in the left anterior descending coronary artery

Blood Vessel

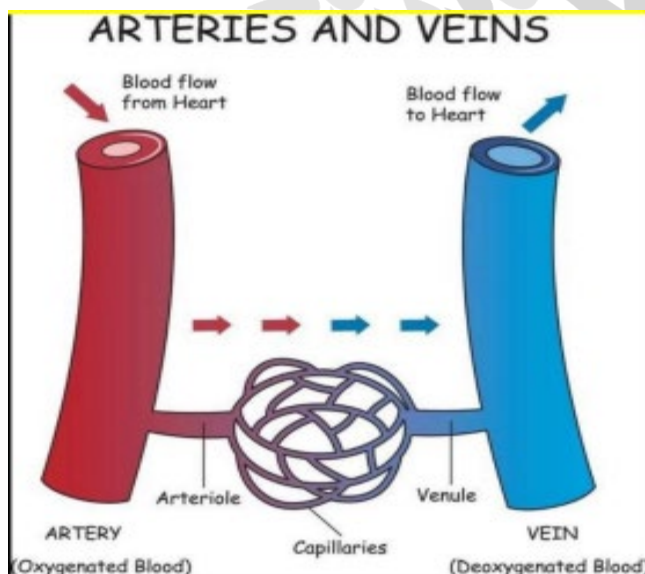
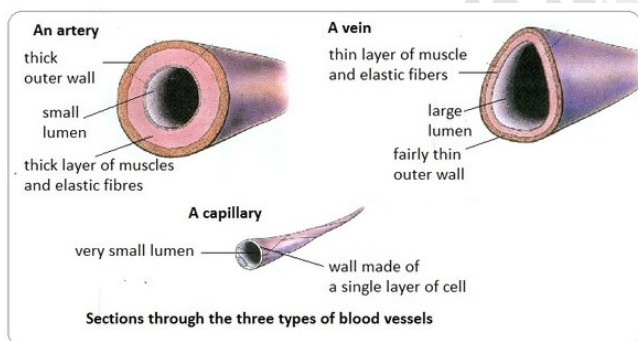
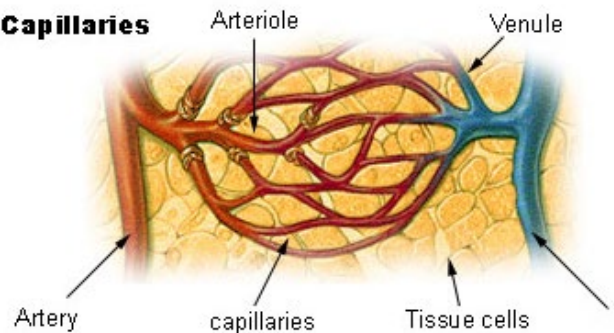
Closed pipe is responsible for transportation of circulating body fluid. Three types of blood vessels are present in human body.

1. Artery
2. Vein
3. Capillary





Artery	Vein
Carries blood away from heart.	Carries blood towards two heart.
Generally, carries oxygenated blood.	Generally, carries deoxygenated blood.
Exception–Pulmonary artery.	Exception–Pulmonary vein.
Narrow Lumen	Wide Lumen
Valves are absent	Valves are present
Blood flow with high speed	Blood flow with slow speed

Capillaries**Capillary**

Capillaries are the body's smallest blood vessels (3–4 microns), acting as the primary site for exchanging oxygen, nutrients, and waste between blood and tissues. They connect arterioles to venules, forming vast networks.

